

1. Use of common global frequencies for all cellular networks
2. Worldwide roaming—anywhere, anytime communication
3. Standardisation of radio interfaces
4. High data transmission rates for both circuit- and packet-switched data
5. Efficient spectrum utilisation schemes
6. Full man and machine mobility
7. Personal communication network (PCN) with UTN

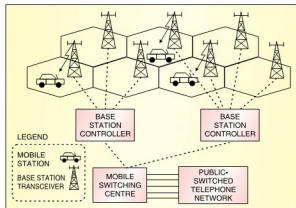
Cellular system architecture

A cellular radio network consists of:

1. **Mobile station.** A device used to communicate over the cellular network
2. **Base station transceiver.** A transmitter/receiver used to transmit/receive signals over the radio interface section of the network
3. **Mobile switching centre.** The heart of the network which sets up and maintains calls made over the network
4. **Base station controller.** Controls communication between a group of base station transceivers and a single mobile switching centre
5. **Public-switched telephone network.** Land-based section of the network

The base station transceivers and base station controller are often collectively known as the base station sub-system.

Cells—the basic geographic unit. A geographic region is divided into cells. Each cell has a base station transceiver that transmits data via a radio link to mobile stations within the cell. A group of base station transceivers is connected to a base station controller. A group of base station controllers is, in turn, connected to a mobile switching centre via microwave links or telephone lines. The mobile switching centre connects to the public-switched telephone network, which switches calls to other mobile stations or land-based telephones.



A cellular system

Coverage areas of different cells

Based on the geographic coverage area, cells in a cellular radio network are classified as macro cells, micro cells, pico cells and femto cells.

Macro. A macro cell provides the largest coverage area within a mobile network—perhaps an entire metropolitan area of 80.5 km (50 miles) in diameter. It is larger than a micro cell and much larger than a pico cell.

But, macro cells provide radio coverage over varying distances depending on the frequency used, the number of calls made and the physical terrain. The antennae for macro cells are usually mounted on ground-based masts, rooftops or other existing structures and are positioned at a height that is not obstructed by terrain or buildings.

Typically, macro-cell base stations have a power output in tens of watts. Macro cells are used to cover the widest range of cell sizes. These are found in rural areas or along highways.

Micro. Micro cells are used over a smaller cell area such as in a densely populated urban area. These enable greater frequency reuse by allowing radio frequency propagation to be confined to a small local area.

A micro cell is served by a low-power tower, covering a limited area such as a small colony, mall, hotel or transportation hub. It is larger than a pico cell in both physical size and coverage area, though the distinction is not always clear. Typically, the range

of a micro cell is less than 1.6 km (a mile).

Usually, micro cells are used to enhance coverage in a small but important area or to add network capacity in areas with very dense phone usage, such as train stations.

Pico. Pico cells are used in areas smaller than micro cells, such as part of a building, a street corner or an

airplane cabin. The pico cell is the smallest of the cells in a cellular personal communications system (PCS) or PCN. It is most commonly used in connection with 3G systems.

A cellular base station designed to serve a relatively small area, such as a single building or a city block, is the basic motive of pico cells. Usually, pico cells are used to extend coverage to indoor areas where outdoor signals do not reach well or to add network capacity in areas with very dense phone usage, such as train stations. Pico cells may combine the electronic equipment and antenna in one integrated unit.

Femto. Currently, the smallest area of coverage that is proposed to be implemented is with a femto cell. Femto cells are proposed to be used in homes or small offices. These are perhaps the most exciting products and challenging technology emerging in the communications market today. According to Dr Alastair Brydon of Analysys, "Femto cells have the potential to transform the telecom industry."

Originally, femto cells were conceived as low-cost consumer products, but today it is becoming clear that these cells will be deployed in a diverse range of communications infrastructure applications. Many researchers believe that femto cells will be a fundamental technology enabler in the deployment of 3G network infrastructure technologies such as UMTS.

"Femto cells are an essential part of operator strategy going forward. pico-